Zeolite Vision: Structural Mapping via SOAP and Principal Components

Abstract

This project leverages data science methodologies, including feature engineering, unsupervised learning, and supervised machine learning, to analyze organic molecules known as Organic Structure-Directing Agents (OSDAs). These agents play a crucial role in guiding the formation of nanoporous materials like zeolites, which are widely used in catalysis, environmental cleanup, and drug delivery. By combining structural analysis and predictive modeling, this study aims to streamline the discovery of efficient and cost-effective materials for applications requiring advanced catalytic or filtration systems.

The project examines the structural diversity and functional properties of OSDAs using advanced molecular descriptors and computational tools. First, Principal Component Analysis (PCA) of Smooth Overlap of Atomic Positions (SOAP) descriptors provides a comprehensive map of structural similarities. K-means clustering and hybridization analysis refine this mapping, revealing distinct groupings and functional patterns. Second, supervised learning methods like Support Vector Machine (SVM) classification are employed to predict the utility of hypothetical OSDAs. The results demonstrate that combining SOAP descriptors with classical molecular descriptors, such as those calculated using Mordred, improves predictive accuracy and provides deeper insights into the structure–property relationships in these complex systems.

Introduction

The development of nanoporous materials like zeolites is a cornerstone of modern industrial processes. Organic Structure-Directing Agents (OSDAs) facilitate the formation of these materials, influencing their pore structure, stability, and catalytic properties. A systematic understanding of the structural and functional diversity of OSDAs is essential for designing advanced materials that minimize energy use and environmental impact.

While traditional descriptors such as distances, angles, and ring sizes offer some insight into OSDA structures, they often fail to capture complex interatomic relationships. Advanced methods like the SOAP (Smooth Overlap of Atomic Positions) descriptor provide a more nuanced representation of atomic environments, enabling better predictions of material properties. Similarly, the use of Mordred descriptors, which quantify a molecule's structural, electronic, and topological features, complements SOAP by capturing global molecular properties.

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This report outlines a two-phase investigation:

1. Summer Work: The SOAP method was used for structural analysis of OSDAs, with PCA applied for dimensionality reduction and visualization.
2. Fall Work: Expanded feature engineering using Mordred descriptors, combined with k-means clustering and hybridization studies, identified key structural patterns. PCA was applied to the combined dataset, revealing top contributing descriptors and enhancing the understanding of structure-property relationships.

Methods

The detailed code and scripts for this analysis can be found in this Github Repository.

1. **Database and OSDA Features Extraction:**
    - Utilized the OSDB (Organic Structure-Directing Agents Database) from the ZeoDB platform to extract relevant features such as SMILES, InChIKeys, volume, host-guest energies, and other structural information for 1194 OSDAs.
    - Database 1 (zeolitesList.csv) : Extracted number of zeolites and a list of zeolites produced by each SMILES structure in the OSDB.
    - Database 2 (input_smiles.csv): Used RDkit to convert SMILES structure to xyz coordinates for each OSDA
        - RDkit could not recognize SMILES structure for the following 7 molecules hence faced error retrieving XYZ for the same -
            - `CC[N+]12C[N@]3C[N@@](C1)C[N@](C2)C3`
            - `C[C@H]1CC[N+](C)(C)[C@@H]2C[C@@H]1C2(C)C`
            - `C[C@@H]1CC[N+](C)(C)[C@H]2C[C@H]1C2(C)C`
            - `CC1(C)[C@@H]2CCC[N+](C)(C)[C@@H]1C2`
            - `CC1(C)[C@H]2CCC[N+](C)(C)[C@H]1C2`
            - `CC[N@+]1(C)CC[C@H](C)[C@@H]2C[C@@H]1C2(C)C`
            - `CC[N+]1(CC)CCC[C@H]2C[C@H]1C2(C)C`
        - Now working with 1194 - 7 = 1187 molecules only.

2. **SOAP (Smooth Overlap of Atomic Positions) Descriptor Calculation:**
    - Employed the SOAP method to generate feature representations for each structure.
        - SOAP Parameters: Radial functions: 8 (max radial = 8); Angular functions: 7 (max angular = 6)
        - Refer to soapCalculation.py in the github repo, pay attention to comments for important tasks.
        - SOAP represents the atomic environment as a three-body correlation function based on the atomic density centered at each atom.

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- Calculated SOAP vectors with a carbon-centered and nitrogen-centered approach.
    - Radial cutoff radii used: 2Å, 4Å, and 6Å.
    - 6 new databases: DatabaseSOAP-X-R where X=C or N, R = 2, 4, or 6.
    - Calculated environment and Molecule SOAP, the latter is averaged over relevant atoms in Molecule `[refer lines 73-77 in soapCalculation.py]`
    - Output files - `smiles_and_soapVectors.csv` and `avg_soap_RX.csv` are too big to download and view so should be viewed on skywalker using vim [filename]

3. **Dimensionality Reduction with PCA:**

- Performed Principal Component Analysis (PCA) on SOAP vectors to reduce the dimensionality of the data.
    - Refer to [pcaCalculation.py]() in the github repo. Make sure to use the correct input file.
- Standardize by shifting mean to zero and scaling SD to unity, for each element of the database of SOAP vectors for Database-X-R.
- Focused analysis on the first two principal components- PC1 and PC2
- Processed molecular data to identify and analyze structural features that are relevant for research and writing the results to a json file.
    - Refer to [pca_chemiscope.ipynb]() for more details and steps.
- Using [chemiscope]() and the json file generated, plotted PCA graphs for visualization and further analysis.

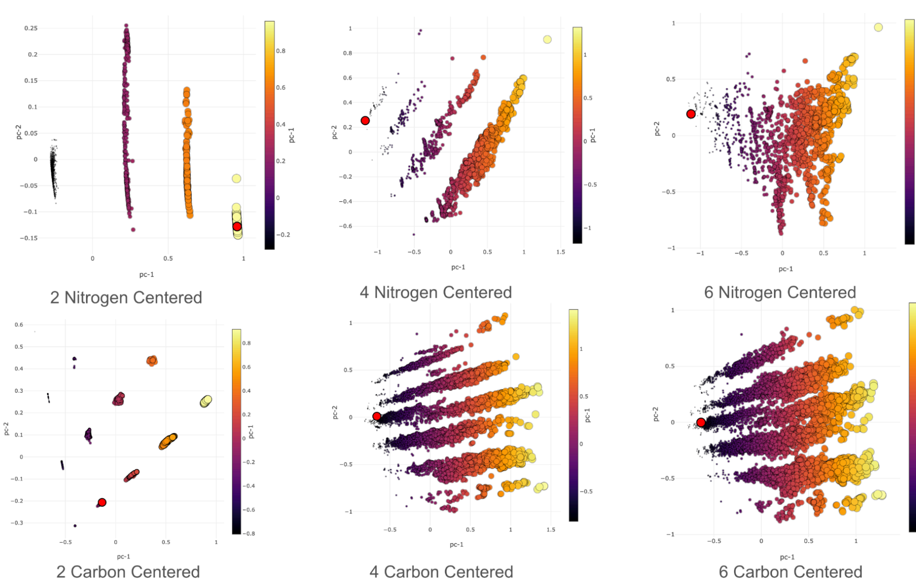

Figure 1: PCA Graphs for Environment System

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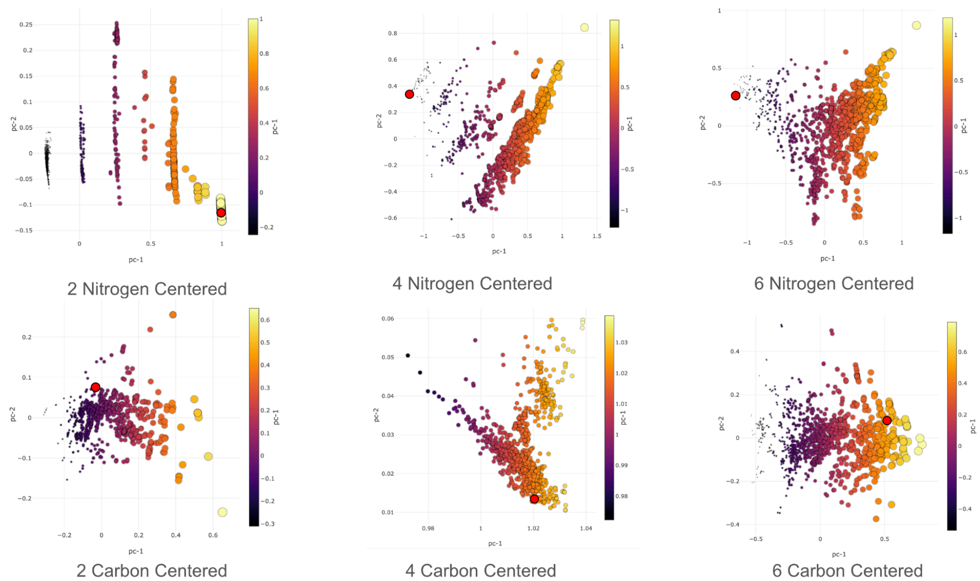

Figure 2: PCA Graphs for Molecule System

- From figure 1 and figure 2, it's clear that 4 Nitrogen-Centered and 4 Carbon-Centered PCA plots show clearer separation and clustering of data points compared to other conditions, indicating that these features best capture the essential variance, making them optimal for focused analysis going forward.

4. **Feature and Correlation Analysis:**
    - **C-Centered:** Analyzed features such as ring distribution, local density, averaged bond angle, coordination number, and types of bonded elements to center atom.
        - These features were selected based on their strong contribution to the principal components, ensuring a comprehensive understanding of the structural variations.
    - **N-Centered:** Focused on ring distribution, local density, averaged bond angle, and the relationship between the four quantized islands and coordination number.
        - This approach was chosen because the primary variance in the data is likely related to these features, and further analysis is needed to clarify their contributions to the principal components.
    - Using scipy.stats.pearsonr, computed Pearson correlation coefficients between different features to understand the relationships between principal components.
        - Refer to the next section for results.

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5. **Mordred Descriptor Calculation and Clustering**

   In the fall, I utilized the Mordred package to compute 1825 molecular descriptors for each OSDA using the code in Calculating_Descriptors.ipynb and stored the descriptors in file smiles_desc.csv. These descriptors include topological indices, charge distributions, and hybridization states, providing a comprehensive global representation of each molecule's properties. These were clustered using K-means clustering(see Clustering_&_PCA.ipynb)

Discussion

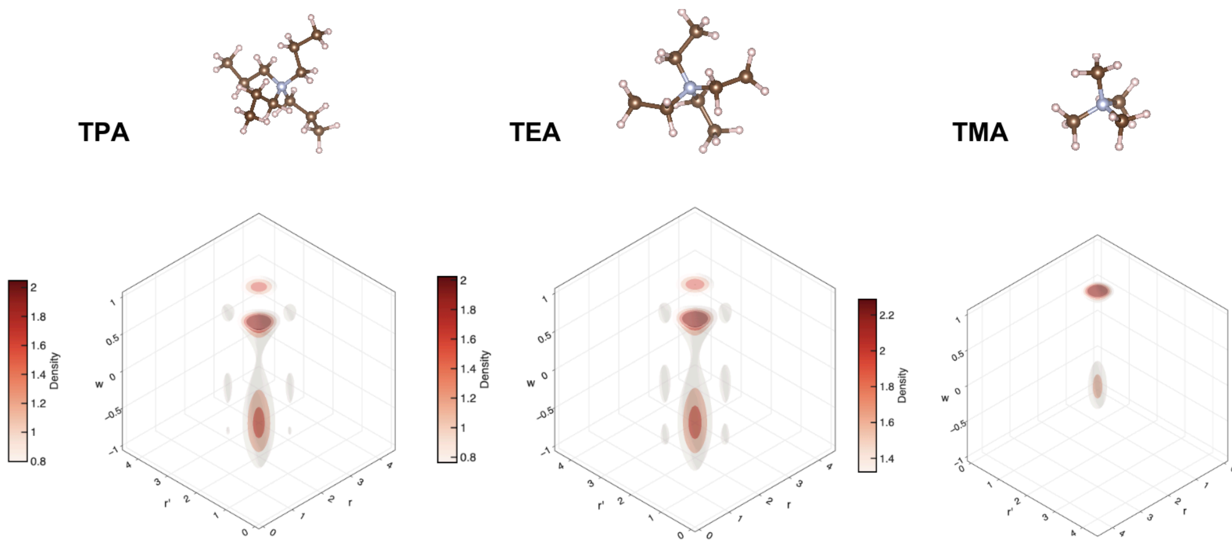

Figure 3: Real-space density plots for TPA, TEA, and TMA molecules

- Figure 3 - We plotted real space density graphs to see how the density distributions in these real-space plots correspond to the expected atomic arrangements around the nitrogen and carbon center. This helped in validating whether the SOAP descriptors are capturing the correct local environments.
  - real_space_power.ipynb includes the steps to generate these density plots.

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Environment

Central atom	Structural Feature (x)	Principal Component (y)	Pearson correlation coefficient (r)
C	Local Density	PC-1	0.824
	Mean Pairwise Distribution	PC-1	0.515
	Coordination Number	PC-2	-0.906
	Types of Bonded Element	PC-2	-0.977
N	Local Density	PC-1	0.862
	Mean Pairwise Distribution	PC-1	0.643
	Coordination Number	PC-1	0.842
	Types of Bonded Element	PC-1	0.835

Figure 4 : Pearson correlation coefficients for different features w/ PC1 & PC2 for environment system

Carbon-Centered Environments

PC-1:

- **Strong Positive Correlation with Local Density (r = 0.824):** PC-1 appears to be significantly influenced by local density, indicating that environments with higher local density contribute more strongly to this principal component. This suggests that PC-1 is associated with the density of neighboring atoms around the carbon center.
- **Moderate Positive Correlation with Mean Pairwise Distance (r = 0.515):** There is also a moderate correlation between PC-1 and mean pairwise distance. This means that as the average distance between neighboring atoms increases, PC-1 also increases, though not as strongly as with local density.

PC-2:

- **Strong Negative Correlations with Coordination Number (r = -0.906) and Types of Bonded Elements (r = -0.977):** PC-2 is negatively correlated with coordination number and the types of elements bonded to the carbon center. A high PC-2 score is associated with environments where the carbon center has fewer neighbors (lower coordination number) and fewer types of bonded elements. This suggests that PC-2 is related to environments with less complexity in terms of bonding and coordination around the carbon center.

Nitrogen-Centered Environments

PC-1:

- **Strong Positive Correlations with Local Density (r = 0.862), Coordination Number (r = 0.842), and Types of Bonded Elements (r = 0.835):** For nitrogen-centered environments, PC-1 is positively correlated with all three features: local density, coordination number, and types of bonded elements. This indicates that higher PC-1 scores correspond to environments with greater local density, more neighbors (higher coordination number), and a greater variety of bonded elements. PC-1 in nitrogen-centered environments captures the complexity and density around the nitrogen center.

Molecule

Central atom	Structural Feature (x)	Pearson correlation coefficient w/ PC-1	Pearson correlation coefficient w/ PC-2
C	Number of Rings	-0.737	0.169
	Average Ring Size	-0.069	-0.027
	Size of the Molecule	-0.471	0.132
	Averaged Local Density	-0.91	0.23
	Coordination Number	-0.69	-0.65
	Types of Bonded Elements	-0.61	-0.74
N	Number of Rings	0.366	-0.212
	Averaged Local Density	0.88	0.01
	Types of Bonded Elements	0.81	-0.56
	Coordination Number	0.81	-0.56
	Average Ring Size	0.237	-0.047
	Size of the Molecule	0.389	-0.213

Figure 5 : Pearson correlation coefficients for different features w/ PC1 & PC2 for molecule system

Carbon-Centered Molecules:

- **PC-1's Strong Negative Correlation with Mean Local Density (r = -0.91):** For carbon-centered molecules, PC-1 shows a strong negative correlation with mean local density, suggesting that environments with lower overall density of carbon-centered environments are associated with higher PC-1 scores.
- **PC-2's Moderate Negative Correlation with Coordination Number (r = -0.65) and Types of Bonded Elements (r = -0.74):** PC-2 continues to exhibit negative correlations with both coordination number and types of bonded elements, though these correlations are less pronounced than in environments, reflecting a similar trend but with less intensity.

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Nitrogen-Centered Molecules:

- **PC-1's Strong Positive Correlation with Local Density (r = 0.88):** Averaging over all local densities for nitrogen-centered molecules, PC-1 continues to show a strong positive correlation with local density, reinforcing the idea that higher local density is strongly associated with higher PC-1 scores.
- **PC-2's Weaker Correlations:** The generally weaker correlations of PC-2 with all features suggest that PC-2 captures less variation in the characteristics of nitrogen-centered environments and molecules, indicating it may be less informative in distinguishing different nitrogen-centered environments compared to PC-1.

In summary, for carbon-centered environments, PC-1 emphasizes local density and distance, while PC-2 reflects less complex environments with fewer bonds and elements. For nitrogen-centered environments, PC-1 captures the overall complexity and density, and PC-2 has less impact, suggesting PC-1 is more influential in describing the characteristics of nitrogen-centered environments.

Key Findings

1. Principal Component Analysis (PCA)

- **Dimensionality Reduction**:
  PCA effectively reduced the dimensionality of SOAP and Mordred descriptors, capturing the most significant variance in the data within the first two components (PC1 and PC2). These components allowed visualization of structural relationships among molecules.
  - **Key Loadings**: The most influential descriptors identified in PCA included **Zagreb1**, **nHBAcc (number of hydrogen-bond acceptors)**, and **FCSP3 (fraction of sp3 carbons)**, highlighting their importance in defining molecular diversity.
- **PCA Heatmap**:
  - Descriptor loadings were used to generate a heatmap, showcasing the top five descriptors influencing each principal component.
  - Positive and negative correlations in the heatmap revealed how structural and electronic properties affect variance, providing insights into molecular behavior.

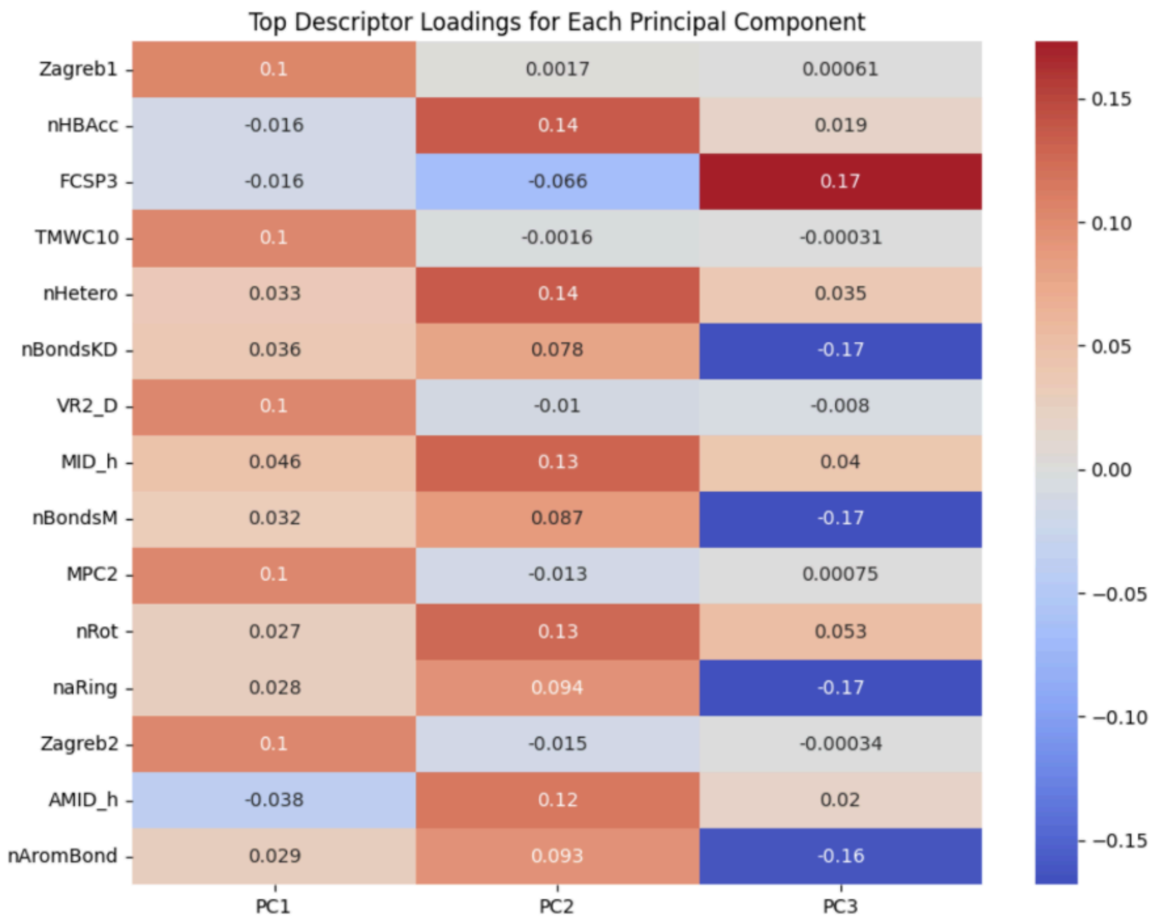

Figure 6: Top Descriptor Loadings for each PC

2. K-Means Clustering

- **Cluster Formation**:
  K-means clustering grouped molecules into three distinct clusters based on PCA-reduced data. This approach captured structural and chemical diversity, with each cluster representing unique molecular characteristics.
  - **Cluster 0**: Defined by topological and electronic descriptors, such as **VE2_D (Vertex Electrotopological state index)** and **Mi (Molecular information index)**. Molecules in this cluster exhibited moderate solubility and aromatic hydrogen influence.
  - **Cluster 1**: Dominated by spatial and spectral descriptors, including **WPath (Wiener Path index)** and **SpAbs_D (Spatial autocorrelation of electronegativity/polarizability)**. These molecules were associated with complex molecular graphs and electronic interactions.
  - **Cluster 2**: Characterized by simpler structural features and drug-likeness properties, such as **GhoseFilter** and **AMID_C (Aromaticity index for carbons)**.
- **Cluster Visualization**:
  Scatter plots of PC1 vs. PC2 scores, colored by cluster, revealed clear separation

between groups, validating the effectiveness of PCA and clustering in capturing chemical diversity.

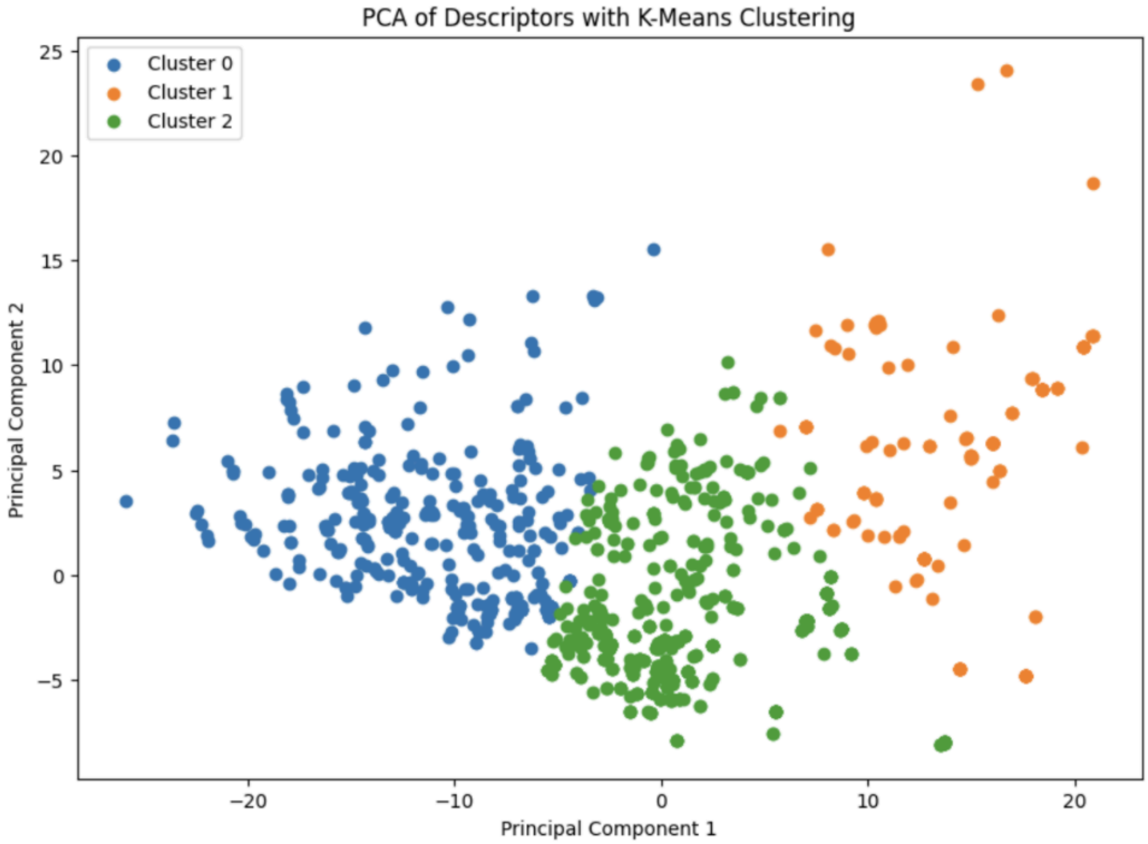

Figure 7: PCA of Descriptors with K-Means Clustering

Top Contributing Descriptors by Cluster:

- Each cluster's defining descriptors reflected unique structural or electronic properties. For example:
  - **Cluster 0** emphasized molecular topology and aromaticity.
  - **Cluster 1** reflected spatial and electronic complexity.
  - **Cluster 2** highlighted simplicity and drug-likeness features.

Hybridization Analysis:

PCA plots revealed distinct patterns of hybridization states, with specific clusters associated with sp2- or sp3-dominated environments. This insight aligns well with the molecular complexity trends observed in descriptors like **piPC7** and **piPC8** (path indices for cyclic structures).

Correlation Heatmaps: Heatmaps of top PCA features revealed strong correlations between local density, coordination number, and molecular descriptors, emphasizing their role in predicting OSDA performance.

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Preliminary SVM Results: Early results suggest that combining SOAP and Mordred features improves classification accuracy for stability and structural diversity.

Summary

We conducted a comprehensive PCA on datasets representing carbon- and nitrogen-centered environments. The analysis focused on identifying and interpreting the principal components (PCs) for each type of center. We evaluated correlations between PCs and various features, such as local density, coordination number, and mean pairwise distance. Additionally, we explored the impact of these features on the principal components and compared the results between environments and molecules. The PCA results reveal distinct patterns in how local structural features influence the principal components for carbon- and nitrogen-centered environments. This study also demonstrates the power of combining SOAP and Mordred descriptors for analyzing the structural diversity of OSDAs. The results highlight the importance of advanced feature engineering and unsupervised learning in understanding complex structure-property relationships. By refining predictive models and expanding the dataset, this approach can facilitate the discovery of more efficient OSDAs, potentially reducing energy costs and environmental impact in material production.

Next Steps -

If time permitted, or if the project were to continue, several additional steps could be undertaken to build on this work:

Supervised Learning and Regression:

- Conduct supervised learning and regression analysis on volume to determine the suitability of different radii and assess the mean absolute error. This will help identify the optimal radius where the learning curve reaches a plateau, indicating whether the SOAP (Smooth Overlap of Atomic Positions) descriptors provide sufficient information.

Database Expansion:

- Enlarge the dataset to include a broader range of organic molecules and non-OSDA (Organic Structure Directing Agent) molecules. This will allow us to evaluate how our findings fit within a larger context and determine if the identified patterns are localized or spread out across different types of molecules.

Acknowledgments

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4. Author links open overlay panelLauri Himanen a, a, b, c, f, d, e, & AbstractDScribe is a software package for machine learning that provides popular feature transformations (“descriptors”) for atomistic materials simulations. DScribe accelerates the application of machine learning for atomistic property prediction by prov. (2019, September 26). *DSCRIBE: Library of descriptors for Machine Learning in Materials Science*. Computer Physics Communications. https://www.sciencedirect.com/science/article/pii/S0010465519303042
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